notebook1

The purpose of this notebook is to document the development of the analysis. The final analysis will be run through a series of scripts, but I will develop those scripts interactively here and document my decision making process as I go.

```
# since this notebook is not in the main directory, use the here package
# to get proper file paths
library(here)
```

```
here() starts at /mnt/c/Users/jjk06/OneDrive/Desktop/2026_r_projs/hybrid_fitness_predict
```

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Seeds were planted 20 at a time into grids of 4 rows by 5 columns. Before planting the seeds, they were placed onto a grid as they were to be planted and a photo taken with a scale in the photo. The areas of the seeds were then measured from the photograph. Four researchers were working on the project, to get used to using imageJ and make sure there was no bias, the seeds were measured by all four researchers.

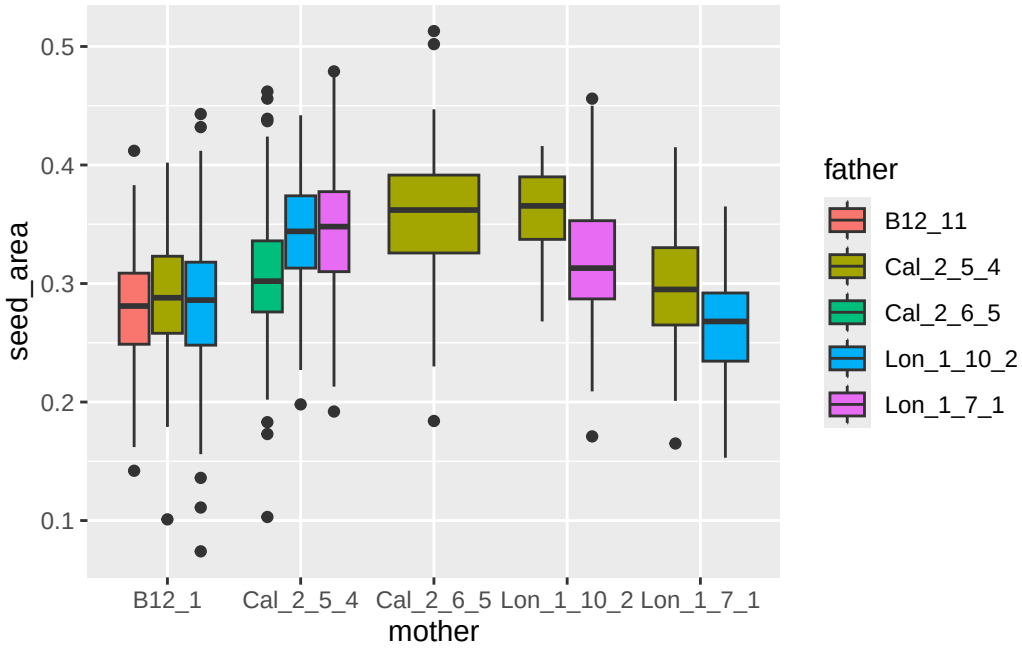
First, I'm reading in the four files and just calculate the mean seed area for each seed.

```r
initials <- c("jk", "js", "lb", "rp")

# there were some notes etc. beyond the 4th column, so only take the first 4 columns
tmp <- read.csv(here(paste0("resources/seed_areas_", initials[1], ".csv")))[,1:4]

tmp$researcher <- initials[1]

d <- tmp

for(i in 2:length(initials)){
  tmp <- read.csv(here(paste0("resources/seed_areas_", initials[i], ".csv")))[,1:4]

  tmp$researcher <- initials[i]

  d <- rbind(d, tmp)
}


# calculate the mean of the four measurements
seed_size <- d %>%
  group_by(tray, block, ind) %>%
  summarise(seed_area = round(mean(area),3))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, and ind.
i Output is grouped by tray and block.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Next, I am going to attach the cross id and parentage to each individual.

```r
ids <- read.csv(here("resources/planting_ids.csv"))

# tray format is different in the two files, so need to change that to merge
ids <- ids %>%
  mutate(tray = gsub("LC", "lc_", tray)) %>%
  select(tray, block, cross, mother, father)
```

```
seed_size %>%
  left_join(ids) %>%
  mutate(cross = tolower(cross),
         cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2", "bc_lon", "lon"))) %>%
  #filter(cross %in% c("cal", "f1", "lon")) %>%
  #mutate(test = paste0(cross, mother)) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area, fill = father)) +
  geom_boxplot()
```

```
Joining with `by = join_by(tray, block)`
```